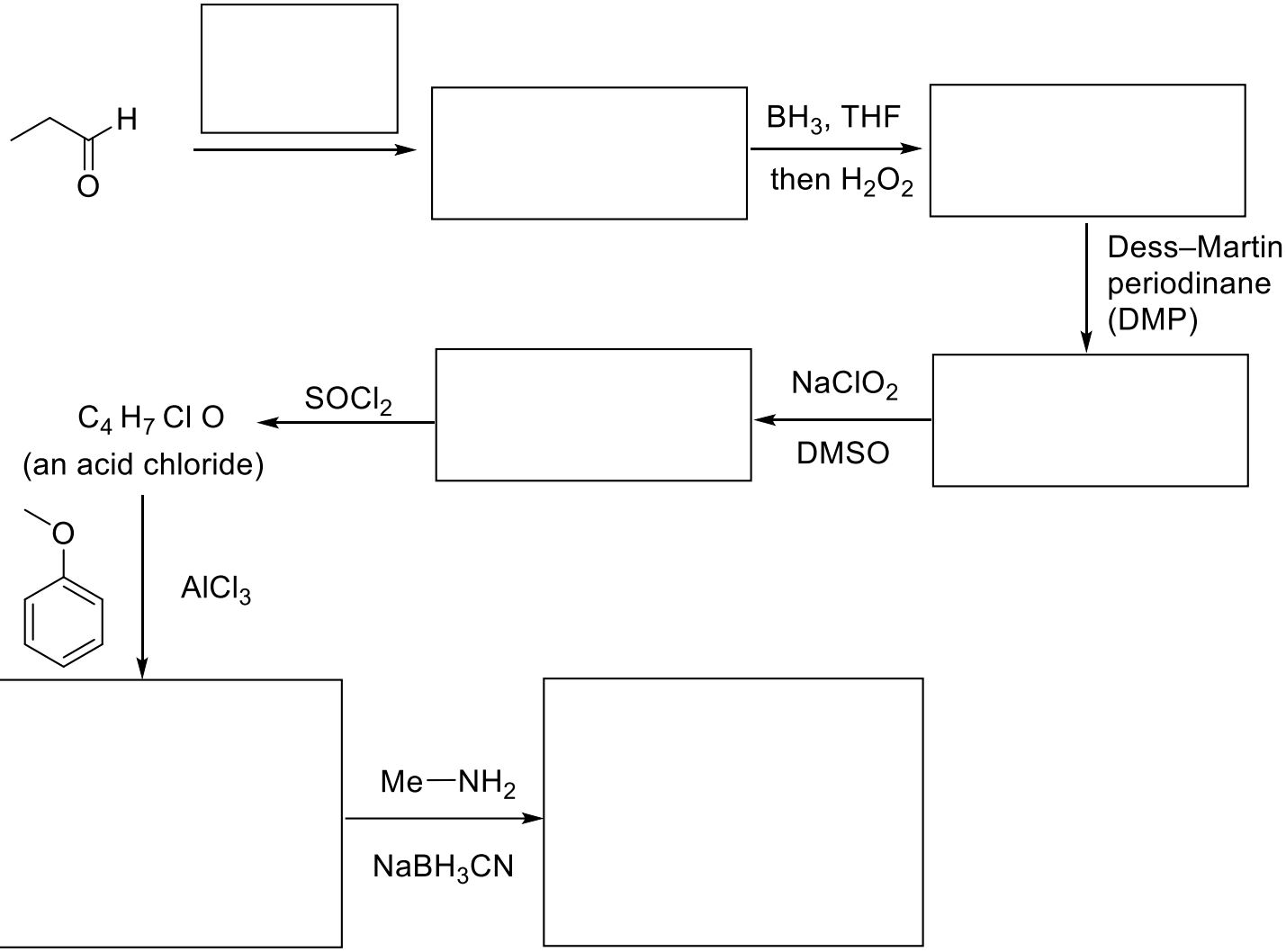
Question 1

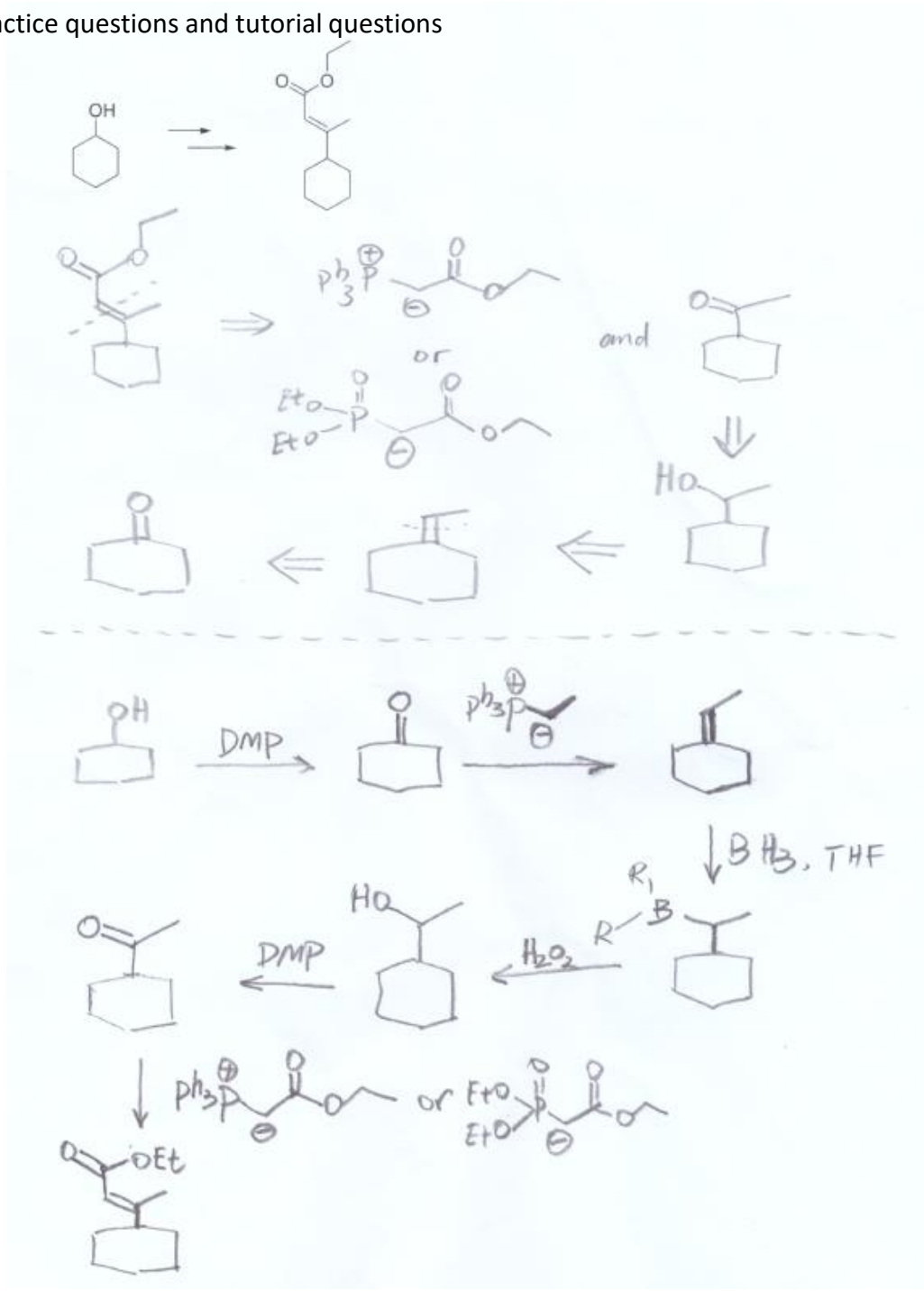
(1) Provide the structures of the reagents, intermediates and products in the boxes. (7 points)

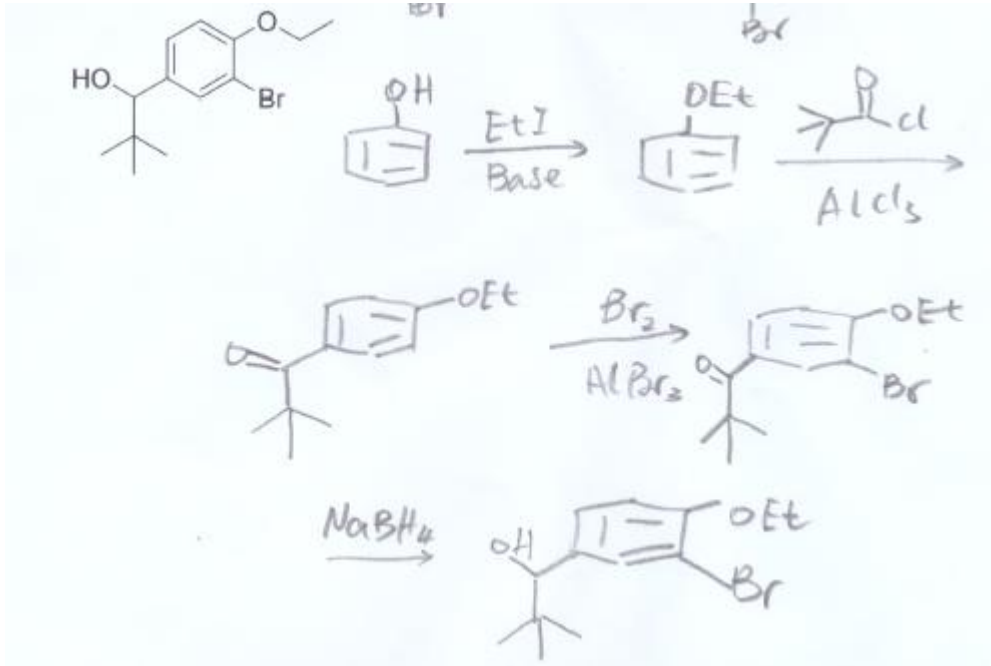


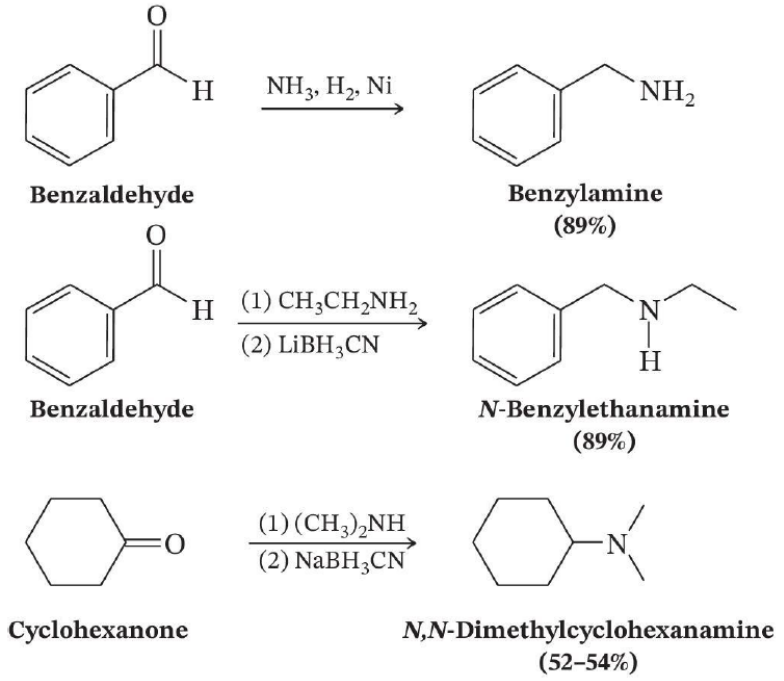
(2) Explain the <u>regioselectivity</u> of the second step (the reaction using BH_3 in THF). (2 point)

(3) Explain why $NaBH_3CN$ works better than $NaBH_4$ in the last step (the reaction with $MeNH_2$). (1 point)

From the practice questions and tutorial questions

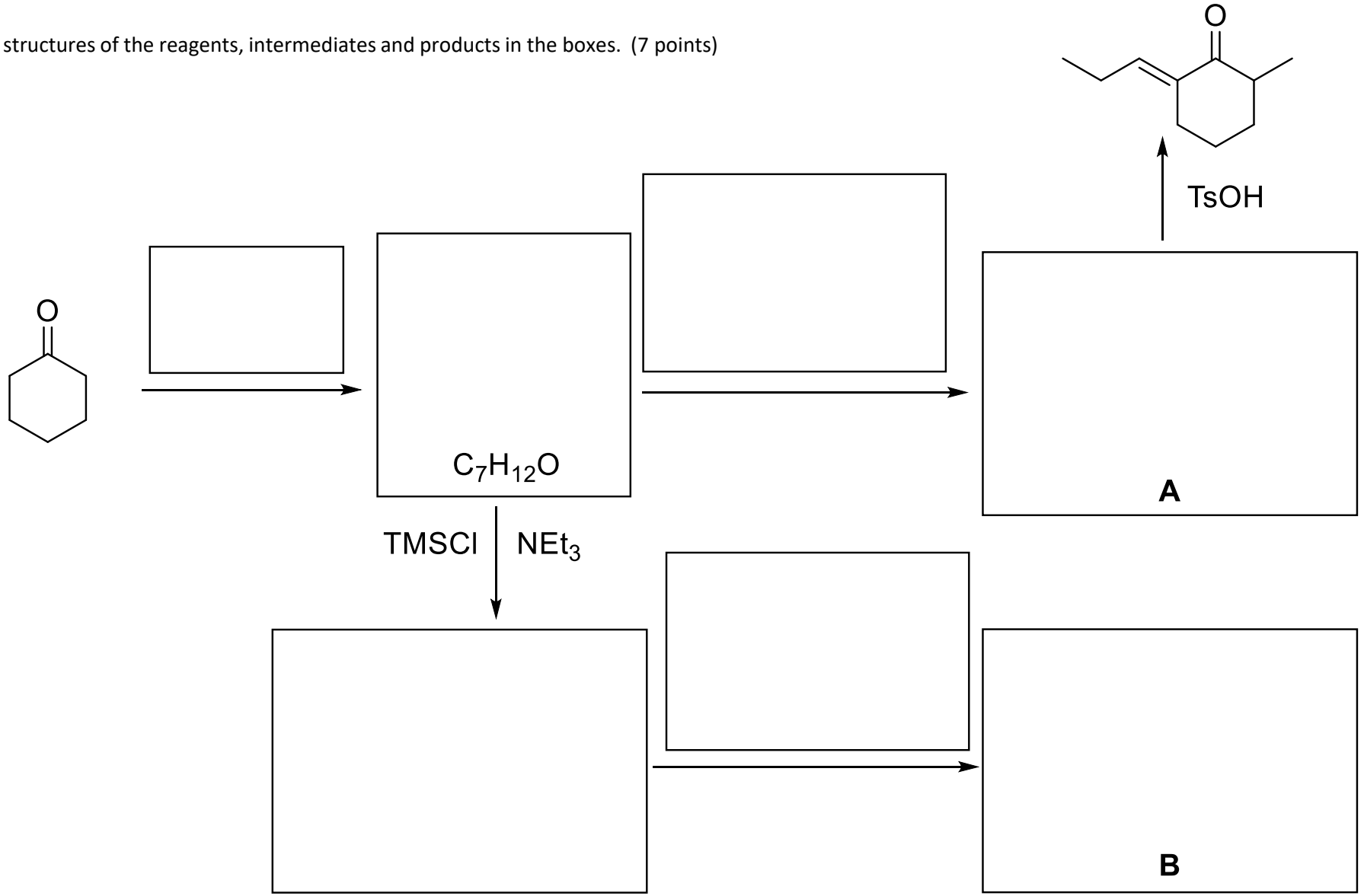






Question 2

(1) Provide the structures of the reagents, intermediates and products in the boxes. (7 points)



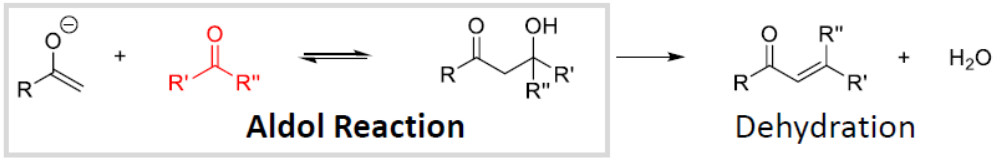
A and **B** are structural (or constitutional) isomers

(2) Explain the regioselectivity for the formation of product **A** and product **B** under the respective reaction conditions that you have provided. (2 points)

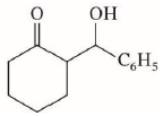
From the practice questions and tutorial questions

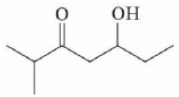
Aldol Reactions

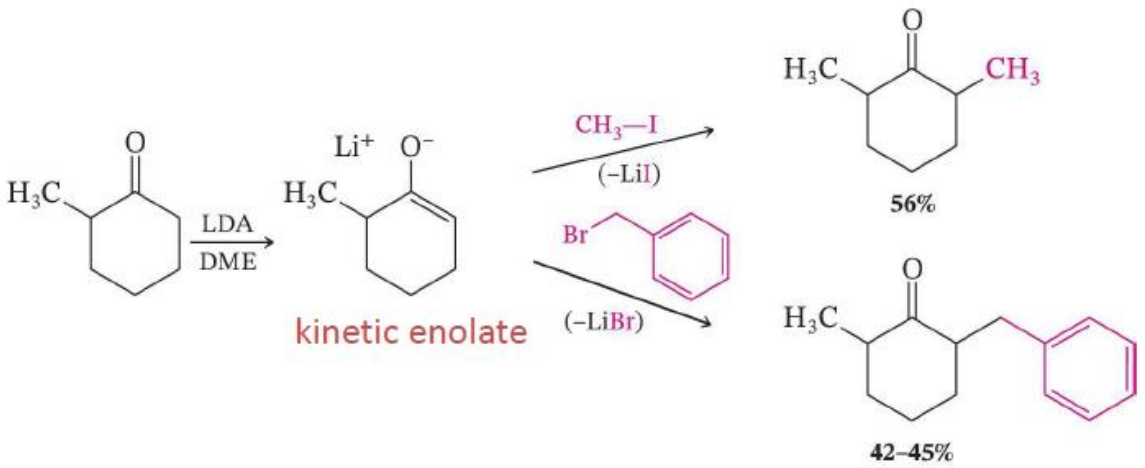
Aldol reactions: Reactions between an enolate (or enol) and a carbonyl compound (aldehyde or ketone)

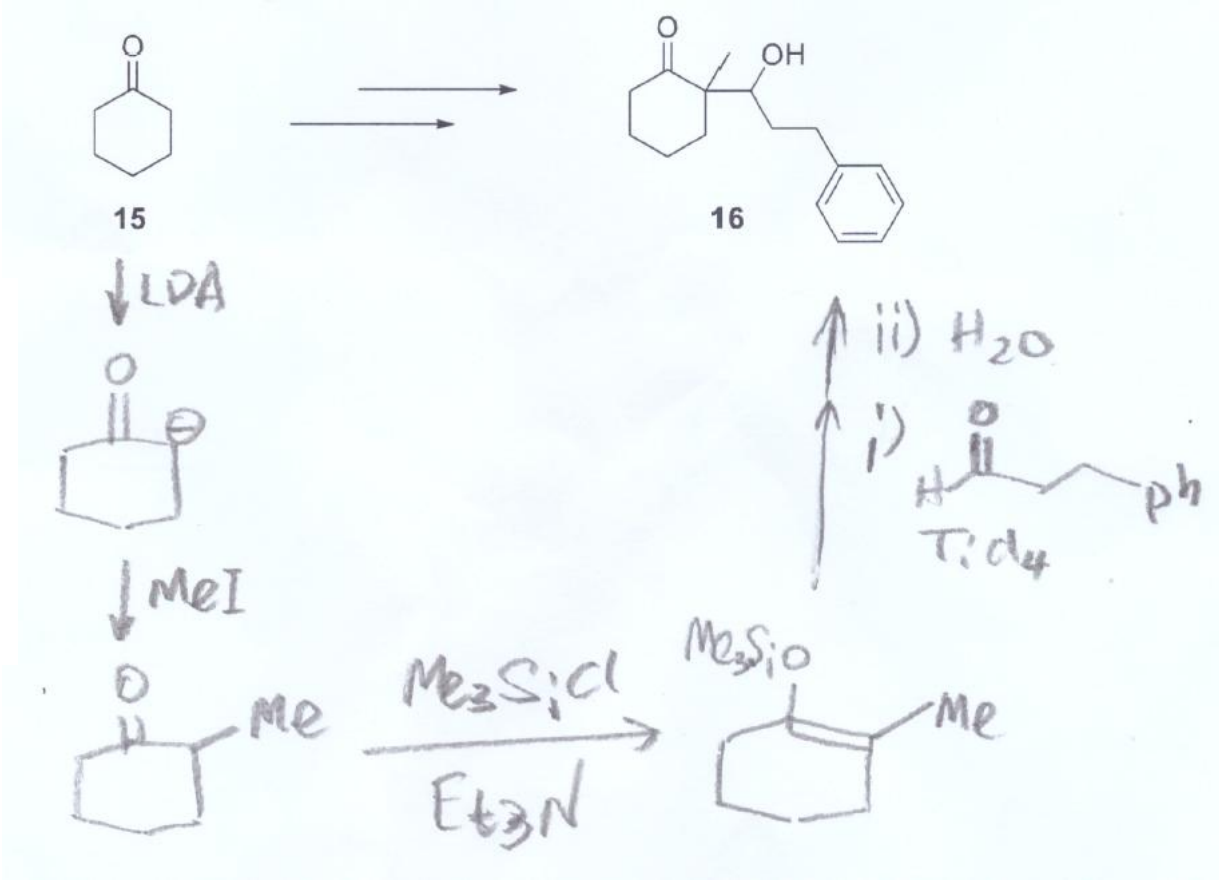


Practice



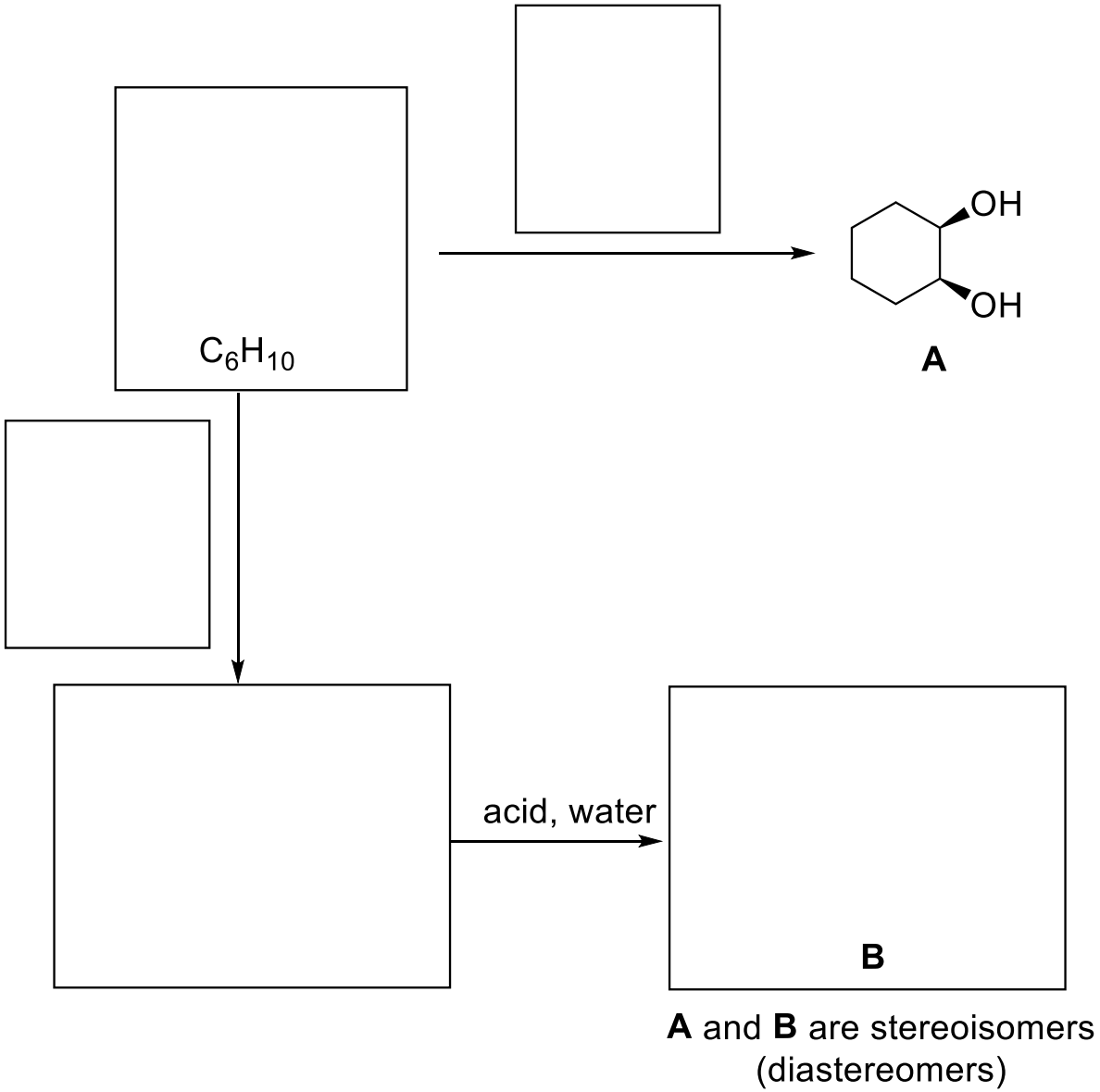






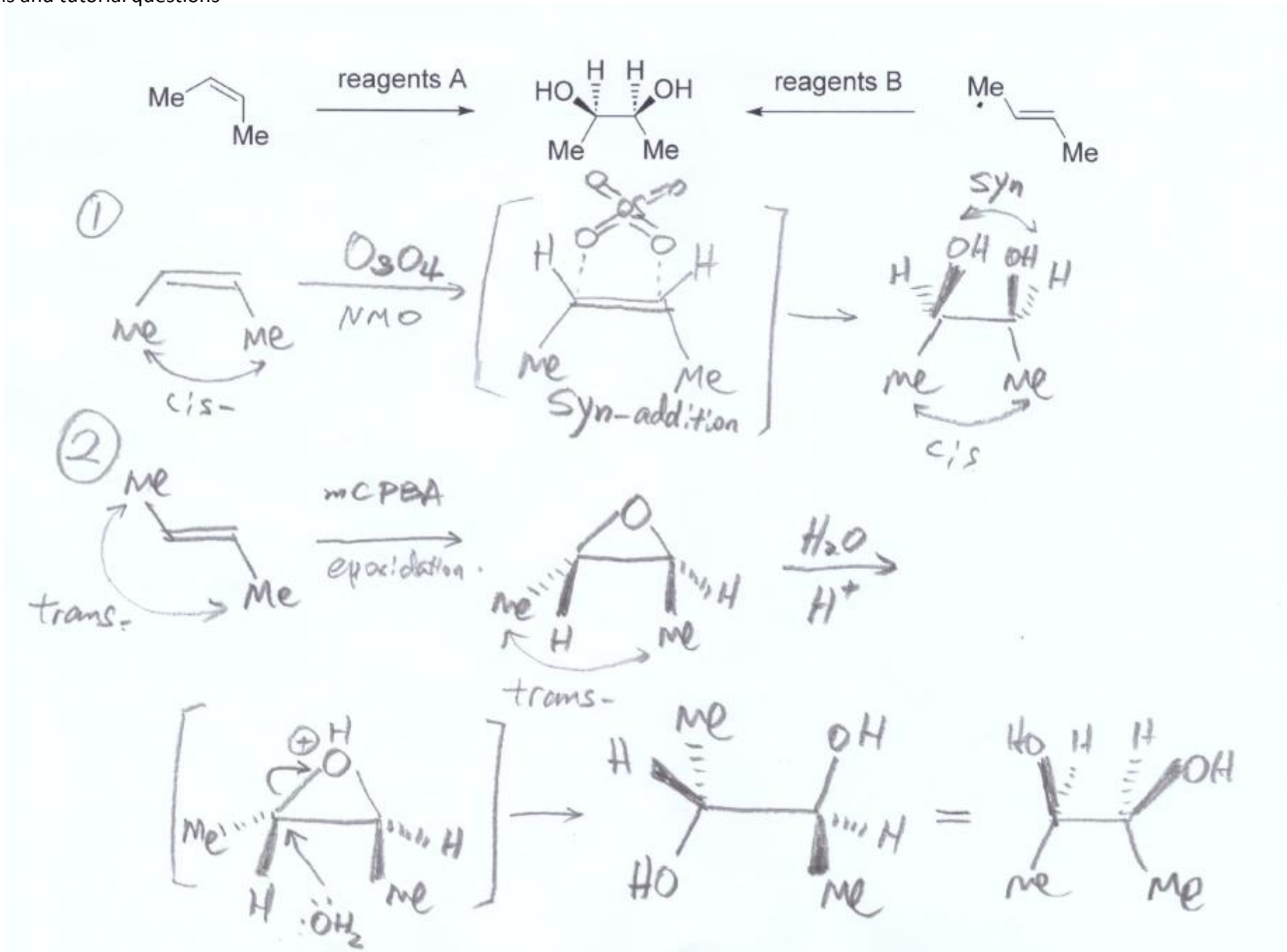
Question 3

(1) Provide the structures of the reagents, intermediates and products in the boxes. (5 points)



(2) Explain the stereoselectivity for the formation of product **A** and product **B** under the respective reaction conditions that you have provided. (2 points)

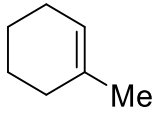
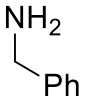
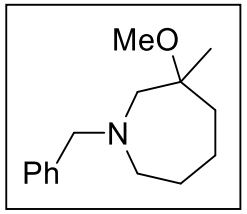
From the practice questions and tutorial questions



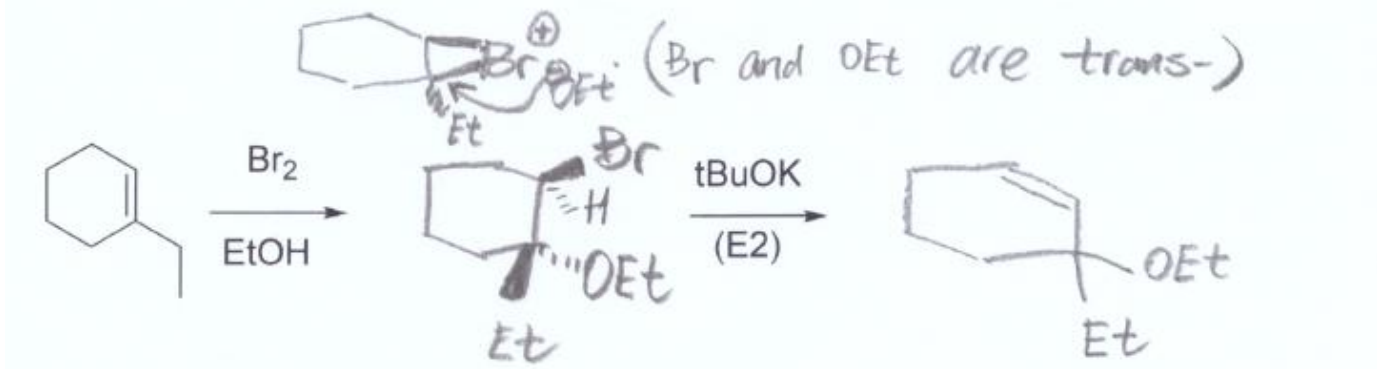
Question 4

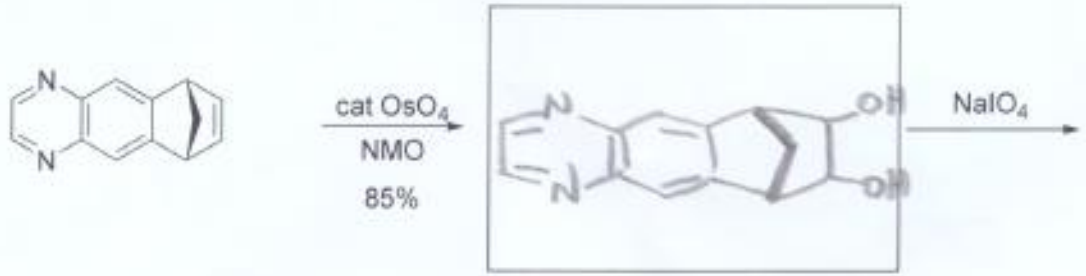
Provide efficient synthetic routes for the following target molecules in the boxes using the starting materials specified. You may also use readily available reagents or reactants as we have discussed in the lectures in addition to these starting materials. Clearly show the products and reaction conditions of each step.

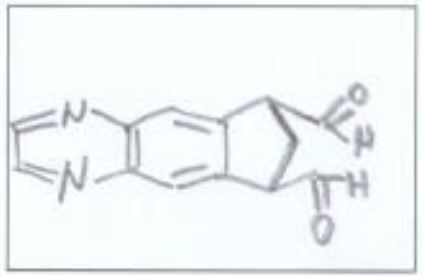
(1) (5 points)

from  and  to 

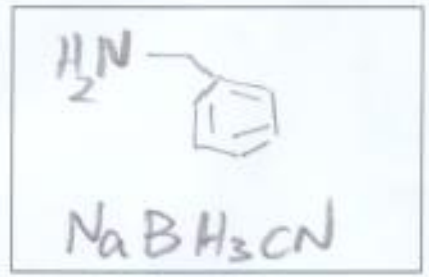
From the practice questions and tutorial questions

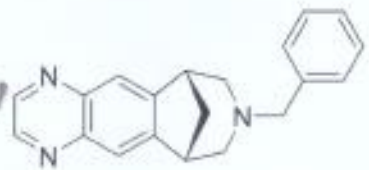






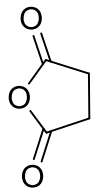
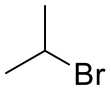
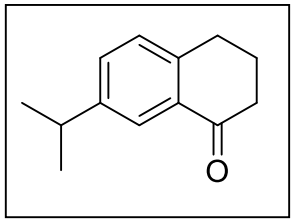
Reductive amination





for Varenicline synthesis

(2) (5 points)

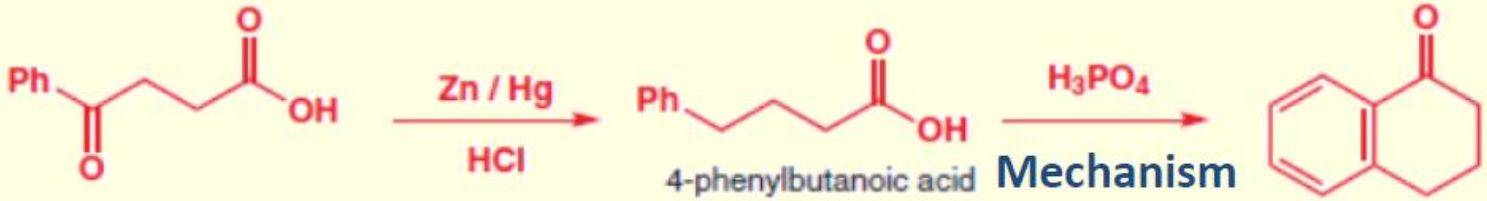
from  and  and  to 

From the practice questions and tutorial questions

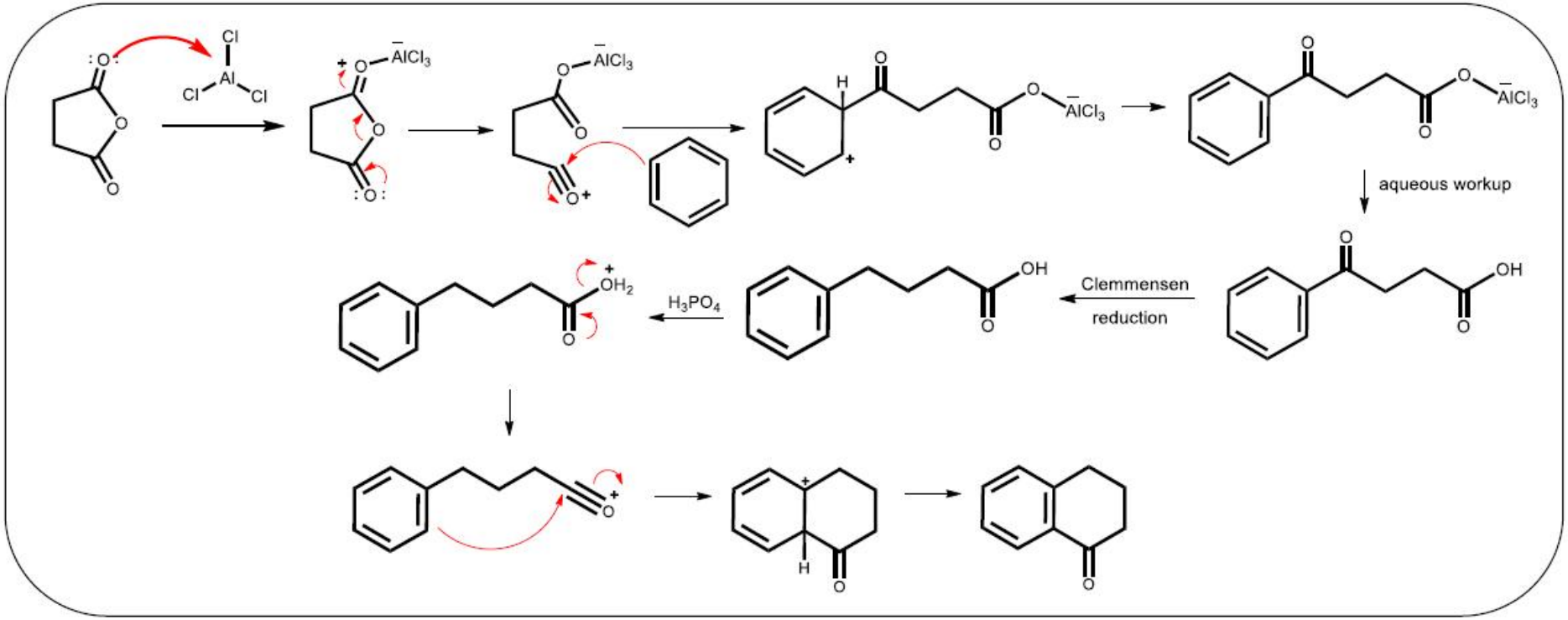
succinic anhydride (the anhydride of butanedioic acid, or succinic acid)



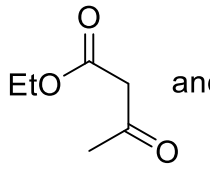
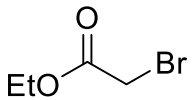
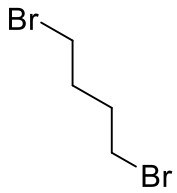
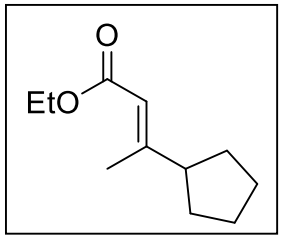
Mechanism



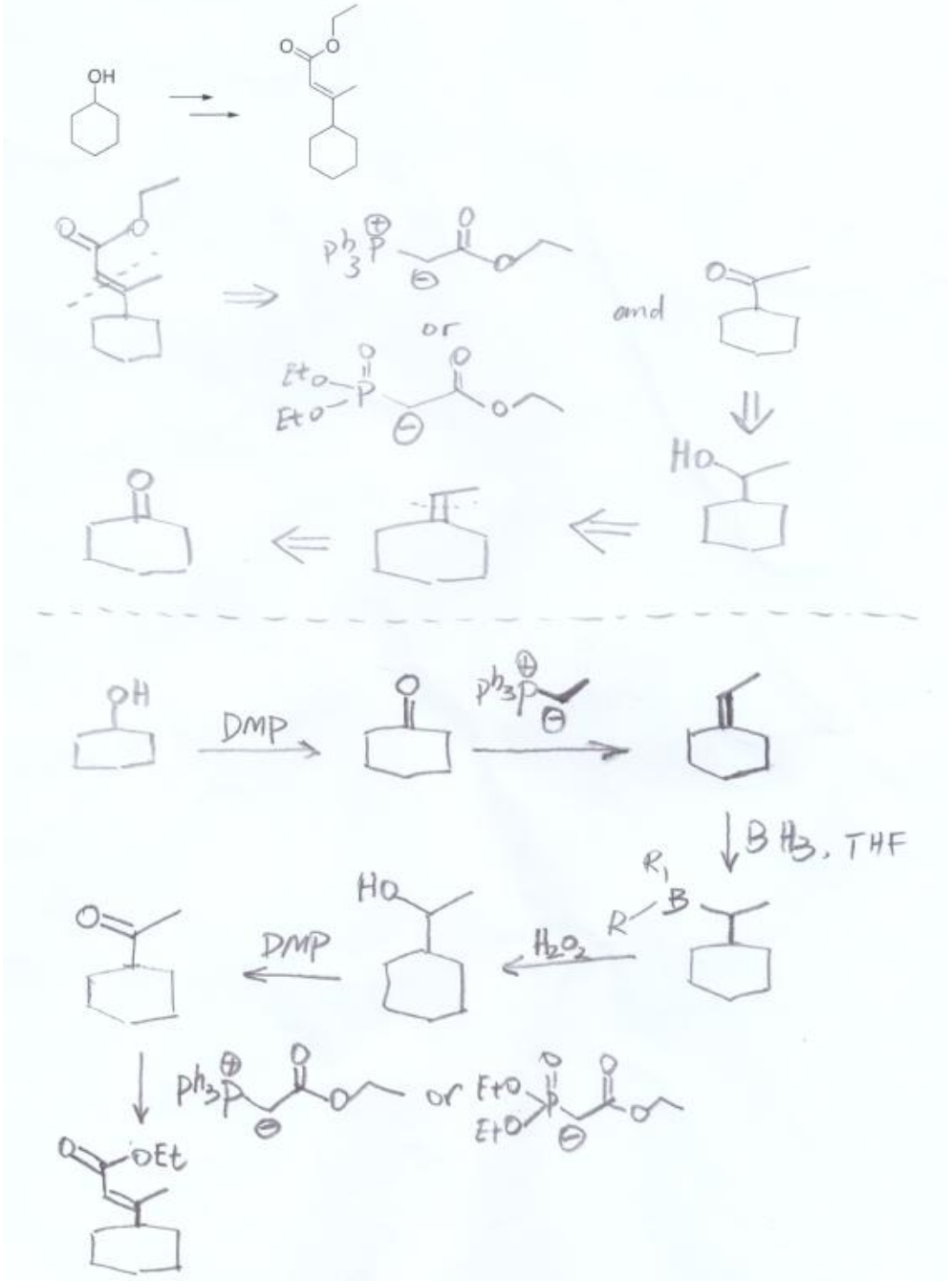
Mechanism



(3) (5 points)

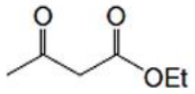
from  and  and  to 

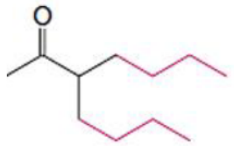
From the practice questions and tutorial questions



Exercise

Synthesis the target molecule starting from ethyl acetoacetate .

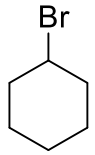


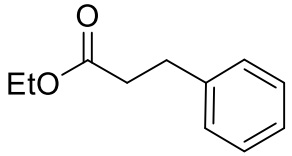


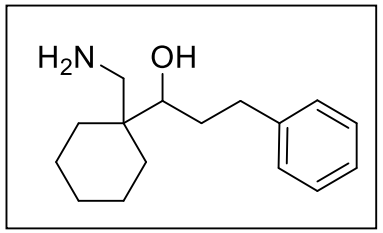
3-Butyl-2-heptanone

Target molecule

(4) (5 points)

from 

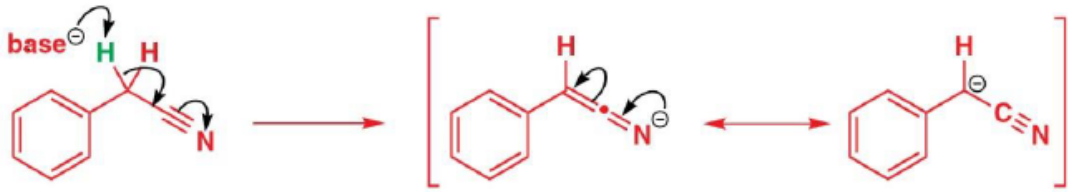
and 

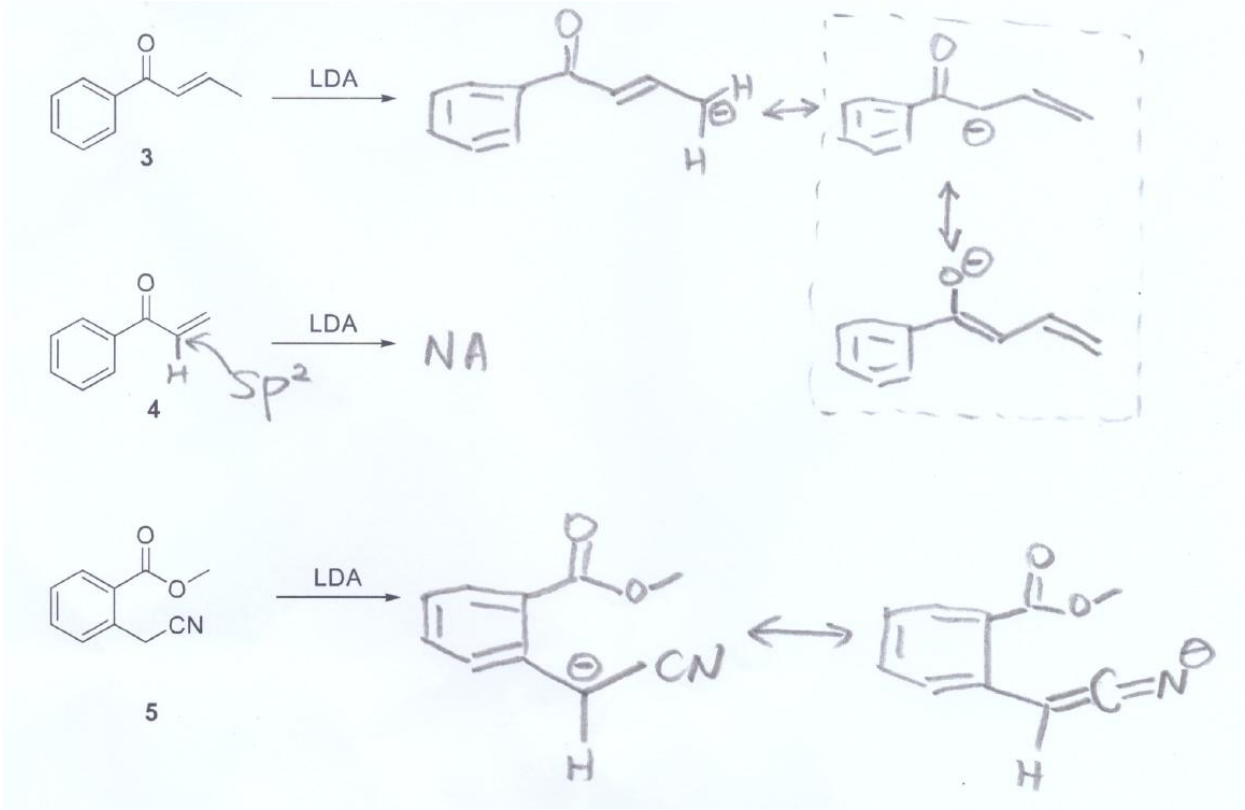
to 

From the practice questions and tutorial questions

Enolate from other functional groups (nitrogen containing)

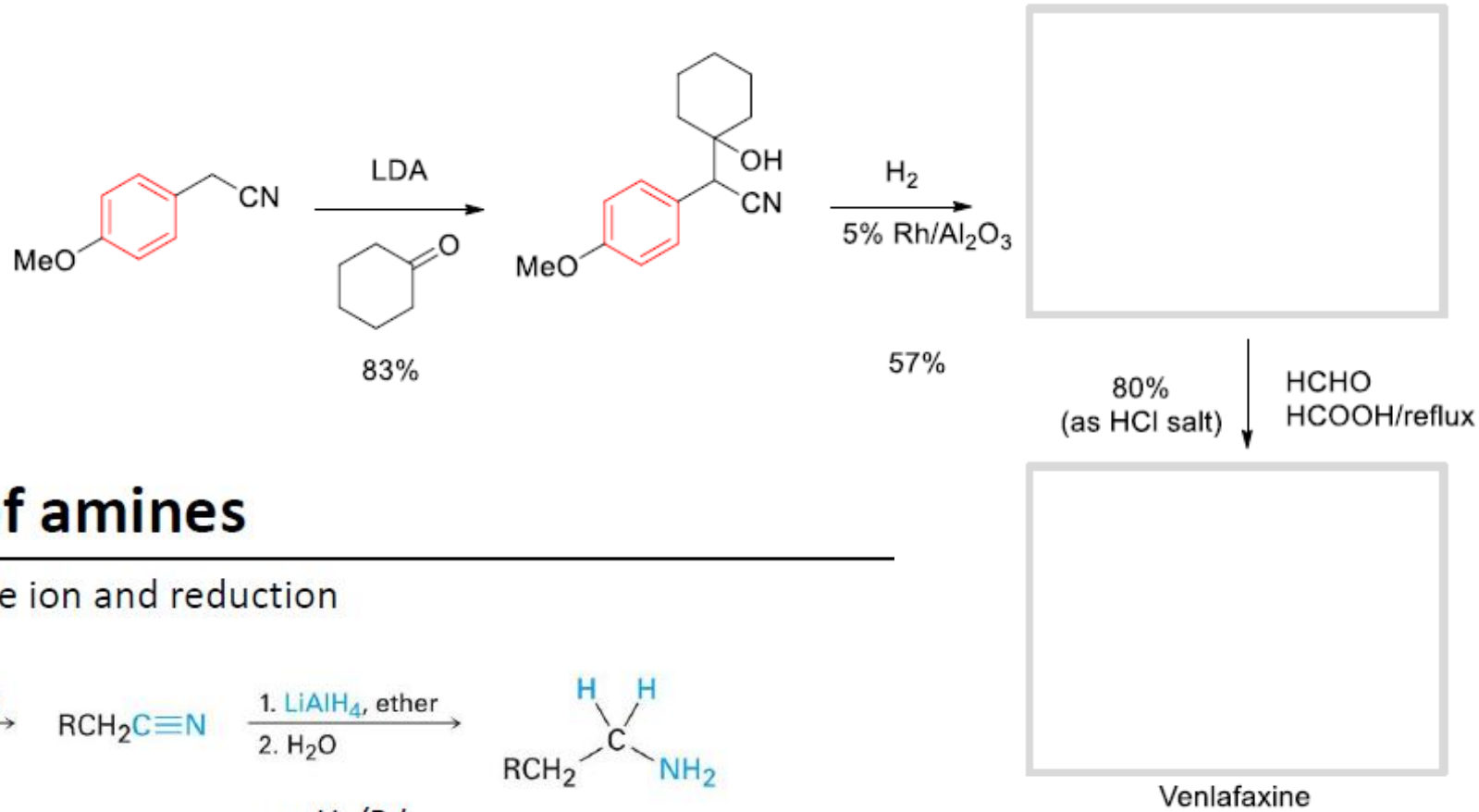
Nitriles





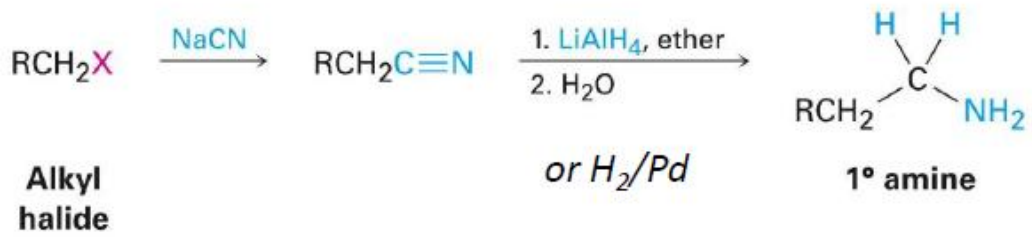
Synthesis of Venlafaxine

Venlafaxine is an antidepressant drug. It is thought to work by increasing the levels of mood-enhancing chemicals called serotonin and noradrenaline in the brain.



Preparation of amines

b. Alkylation of cyanide ion and reduction

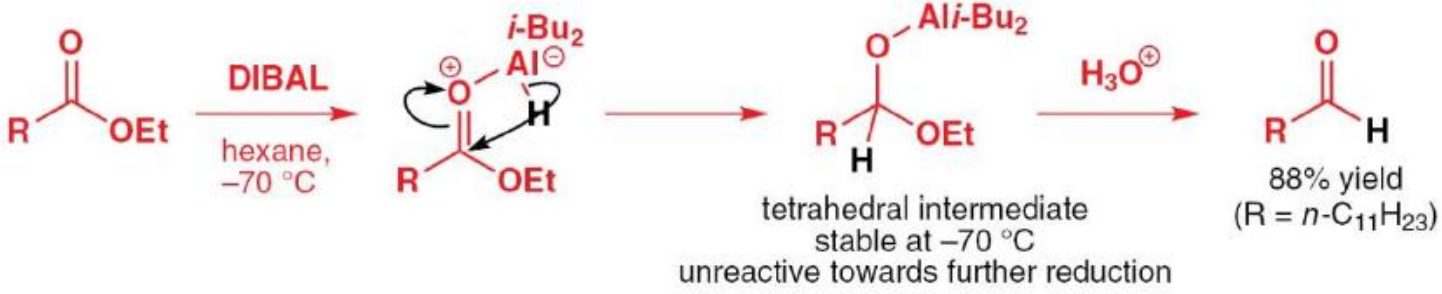


From the practice questions and tutorial questions

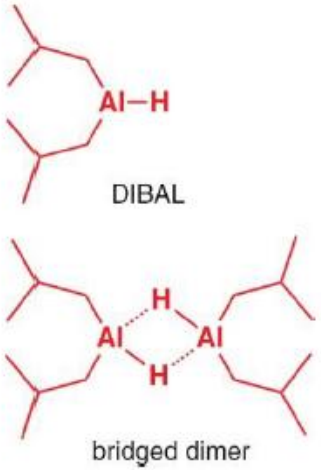
DIBAL-H

DIBAL-H (diisobutyl aluminium hydride, which is an alane):

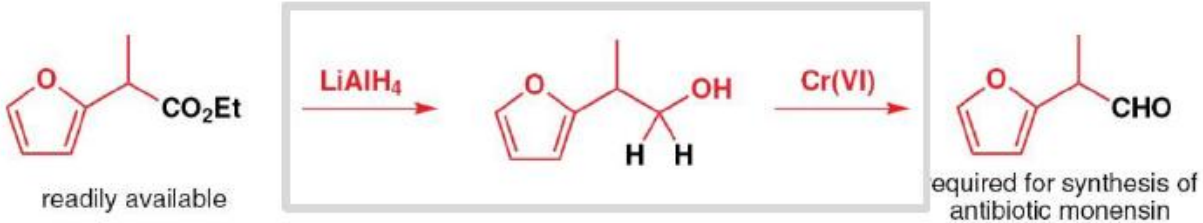
Partial Reduction (esters to aldehyde): at low temperature



tetrahedral intermediate
stable at –70 °C
unreactive towards further reduction



Question: what is the other way to convert esters to aldehydes?



readily available

equired for synthesis of antibiotic monensin

(carboxylic ester→ primary alcohol→ aldehyde)

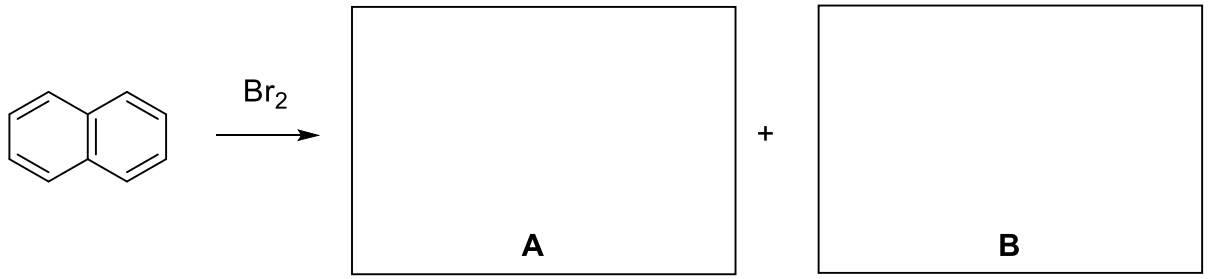


(carboxylic acid→ acid chloride→ aldehyde)

Question 5

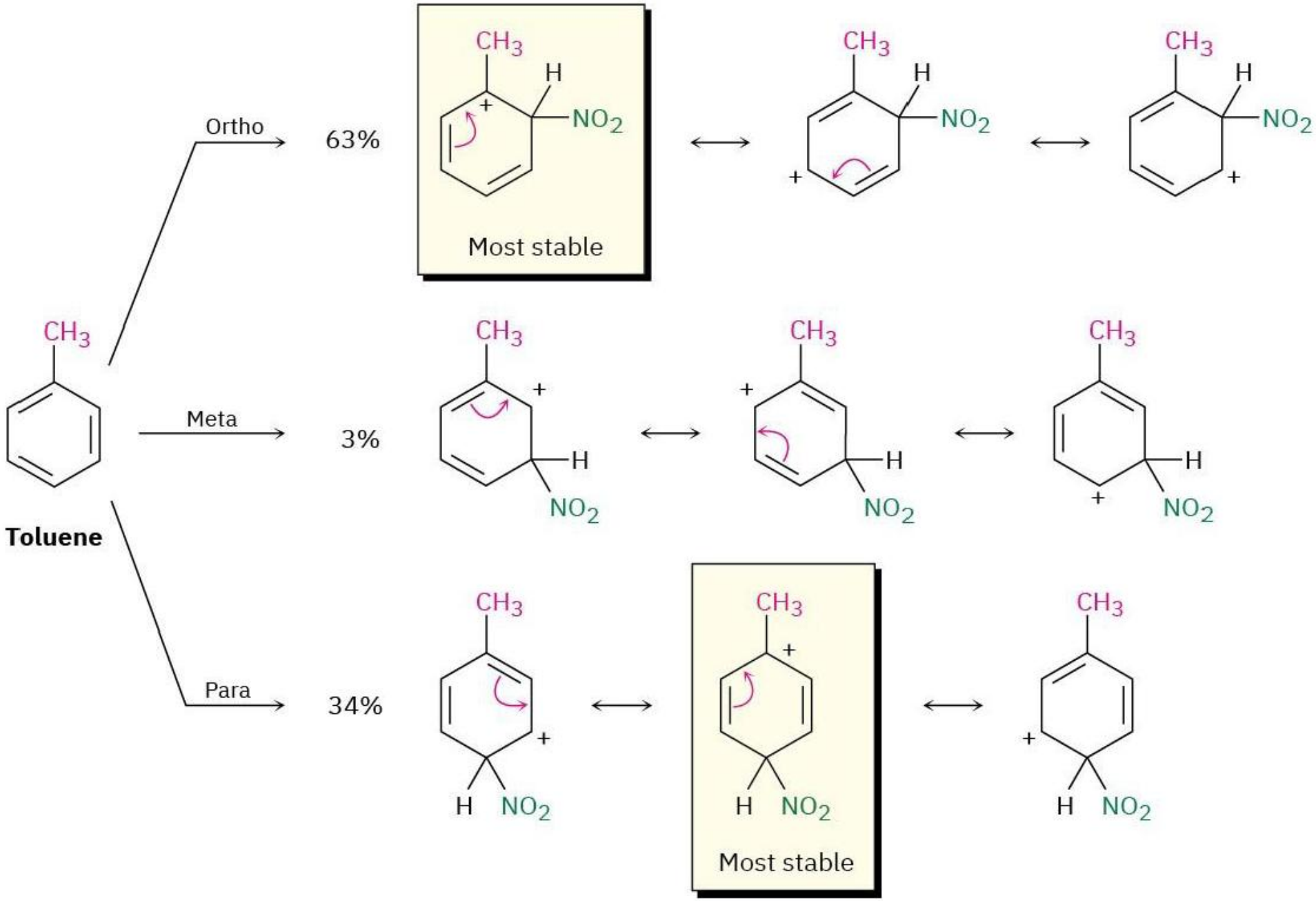
Mono-bromination of naphthalene gives two products **A** and **B** as regioisomers.

•Provide the chemical structures of the two mono-bromination products. (2 points)



•Draw the structures of arenium ions leading to the formation of the *mono*-bromination products. Based on the possible resonance structures of the corresponding arenium ions, which regioisomer is the major *mono*-bromination product? (2 points)

Hint: when drawing the resonance structures, you may assume the "1,3,5-cyclohexatriene" structure of the remaining benzene unit does not change, so that its aromaticity is not disrupted.


Ortho
63%
Most stable
Meta
3%
Toluene
Para
34%
Most stable